

AUTOMOTIVE SOFTWARE SHOULD-COST · MODULE-BY-MODULE DEEP-DIVE

Range Rover L460 software cost deep-dive

A full teardown of all 49 embedded-software modules — features, cost detail, competitive differences and a specific insight for every one — behind the L460 benchmark.

MODULES ANALYSED

49

SOFTWARE DOMAINS

7

OPPORTUNITY

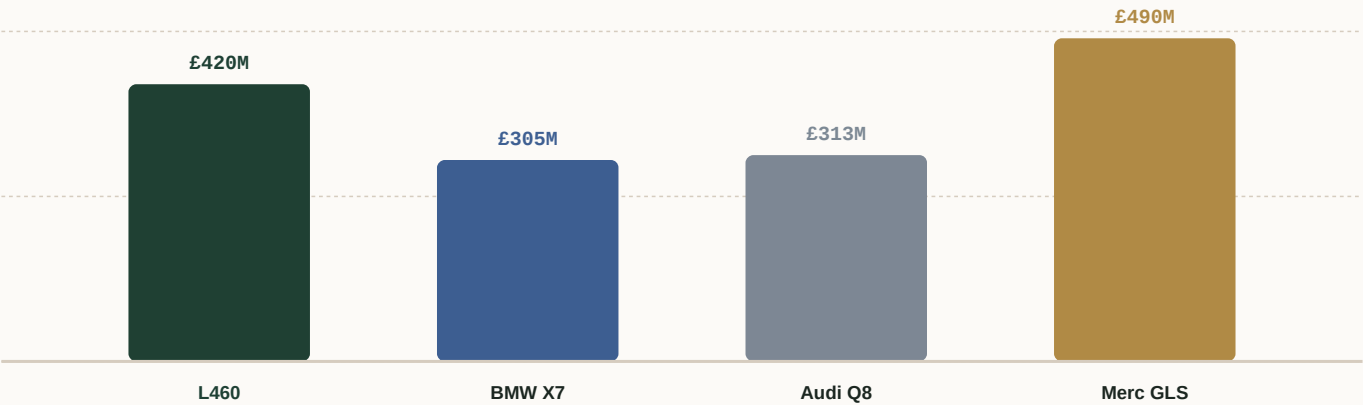
£145M

A The benchmark, in brief

The L460's core mild-hybrid (MHEV) software programme costs materially more than the German rivals it competes with — the only drivetrain they share. The cause is software reuse, and it is addressable. This report then dissects all 49 modules behind that headline.

L460 MHEV PROGRAMME	VS BMW X7 (BEST PEER)	ROOT CAUSE	ADDRESSABLE SAVING
£420.0M	+£114.8M	Lower reuse	£145M
£700 / vehicle	+37.6%	Medium vs Heavy	-35%

MHEV four-way — total software cost (£M)



The gap is not geography or volume (a normalised view keeps the L460 ~£155M above BMW/Audi). BMW and Audi carry far more software forward — "Heavy"/platform reuse (0.35x / 0.14x) vs the L460's "Medium" (0.60x). Two architecture-led levers — Heavy reuse plus shared middleware — take the MHEV programme to ~£275M, a £145M (35%) saving, without offshoring. On electrified programmes the L460 already out-benchmarks the Porsche Cayenne. The following pages price and diagnose every module.

B How to read this section

The L460's software is 49 modules across 7 domains. Each block below states what the software does, its full cost for the L460, the same module priced for the peers that build it, and a specific insight. Mild-hybrid modules are benchmarked against the BMW X7 / Audi Q8 / Mercedes GLS; the seven EV-powertrain modules (which the German MHEVs omit) are benchmarked against the Porsche Cayenne.

The seven domains

A · EV Powertrain & Battery	9 modules	£147.9M	BMS Core · EDU Control · Inverter Ctrl ...
B · ADAS L2 / L2+	9 modules	£191.7M	Camera Perception · Sensor Fusion · Path Planning ...
C · Infotainment & UX	9 modules	£64.8M	IVI OS · Voice Assistant · HMI Framework ...
D · Vehicle Domain Controllers	7 modules	£74.1M	VMM · Chassis Control · Zonal Arch. ...
E · Middleware & Platform	6 modules	£29.6M	AUTOSAR Adaptive · AUTOSAR Classic · Comm. Stacks ...
F · Cybersecurity (ISO 21434)	5 modules	£17.7M	Secure OTA · Secure Boot · IDS / IDPS ...
G · OTA & Cloud Backend	4 modules	£33.5M	Cloud Backend · OTA Manager · Data Pipeline ...

Cost figures are the module's total programme should-cost (NRE + lifecycle) for the L460. "vs best" is the cheapest peer that builds the same module. Savings are the module-level effect of lifting reuse to "Heavy".

BMS Core Software**£42.5M**

ASIL D · Very High · reuse Medium · 173 PM · £71/veh

Battery pack monitoring, protection logic, cell voltage/temp acquisition, state machine management, ASIL-D safety logic.

Dev £9.8M Test £18.7M Integ £1.8M Cyber £1.4M Calib £0.8M Tool £0.4M Maint £9.5M IP £0.1M

VS PEERS Cayenne £48.9M

£-6.4M vs best*Insight — EV-powertrain software the German MHEV peers omit — benchmarked against the Porsche Cayenne. At £42.5M the L460 is £6.4M cheaper than the Cayenne (£48.9M), on its lower UK / Tier-1 blended rate. Heavier carry-forward would save ~£15.7M.***EDU (Electric Drive Unit) Control Software****£32.0M**

ASIL D · Very High · reuse Medium · 125 PM · £53/veh

Integrated electric drive unit control, dual-motor torque vectoring, multi-speed gearbox integration, creep & one-pedal drive.

Dev £7.1M Test £14.1M Integ £1.4M Cyber £1.0M Calib £0.9M Tool £0.5M Maint £6.8M IP £0.2M

VS PEERS Cayenne £36.8M

£-4.8M vs best*Insight — EV-powertrain software the German MHEV peers omit — benchmarked against the Porsche Cayenne. At £32.0M the L460 is £4.8M cheaper than the Cayenne (£36.8M), on its lower UK / Tier-1 blended rate. Heavier carry-forward would save ~£11.8M.***Inverter Control Algorithms****£20.5M**

ASIL D · Very High · reuse Medium · 86 PM · £34/veh

Space Vector PWM, switching frequency optimisation, dead-time compensation, EMI management, demagnetisation protection.

Dev £4.9M Test £9.8M Integ £0.9M Cyber £0.0M Calib £0.5M Tool £0.4M Maint £3.9M IP £0.1M

VS PEERS Cayenne £23.6M

£-3.1M vs best*Insight — EV-powertrain software the German MHEV peers omit — benchmarked against the Porsche Cayenne. At £20.5M the L460 is £3.1M cheaper than the Cayenne (£23.6M), on its lower UK / Tier-1 blended rate. Heavier carry-forward would save ~£7.5M.***Motor Control (FOC/DTC/SVPWM)****£19.2M**

ASIL D · Very High · reuse Medium · 81 PM · £32/veh

Field-oriented control, direct torque control, sensorless rotor position estimation, flux linkage tables, temperature derating.

Dev £4.6M Test £9.1M Integ £0.8M Cyber £0.0M Calib £0.6M Tool £0.3M Maint £3.7M IP £0.1M

VS PEERS Cayenne £22.1M

£-2.9M vs best*Insight — EV-powertrain software the German MHEV peers omit — benchmarked against the Porsche Cayenne. At £19.2M the L460 is £2.9M cheaper than the Cayenne (£22.1M), on its lower UK / Tier-1 blended rate. Heavier carry-forward would save ~£7.0M.***SOC/SOH/SOE Estimation Models****£13.6M**

ASIL C · Very High · reuse Medium · 58 PM · £23/veh

Electrochemical & data-driven (ML) State of Charge, Health, Energy estimation. Kalman, EKF, neural network approaches.

Dev £3.3M Test £4.4M Integ £0.5M Cyber £0.0M Calib £0.4M Tool £0.2M Maint £3.7M IP £0.2M

VS PEERS Cayenne £15.5M

£-1.9M vs best*Insight — EV-powertrain software the German MHEV peers omit — benchmarked against the Porsche Cayenne. At £13.6M the L460 is £1.9M cheaper than the Cayenne (£15.5M), on its lower UK / Tier-1 blended rate. Heavier carry-forward would save ~£4.8M.***Fast-Charging Control Software****£7.3M**

ASIL C · Very High · reuse Medium · 35 PM · £12/veh

CCS/CHAdeMO/OCPP protocol stacks, dynamic power curve management, thermal derating during charge.

Dev £2.0M Test £2.4M Integ £0.3M Cyber £0.2M Calib £0.2M Tool £0.1M Maint £1.9M IP £0.1M

VS PEERS Cayenne £8.3M

£-1.1M vs best*Insight — EV-powertrain software the German MHEV peers omit — benchmarked against the Porsche Cayenne. At £7.3M the L460 is £1.1M cheaper than the Cayenne (£8.3M), on its lower UK / Tier-1 blended rate. Heavier carry-forward would save ~£2.7M.*

Battery Thermal Management Software

£4.4M

ASIL B · High · reuse Medium · 32 PM · £7/veh

Thermal control loops, coolant pump/valve actuation, fast-charge thermal preconditioning, cabin integration.

Dev £1.5M Test £1.2M Integ £0.2M Cyber £0.0M Calib £0.1M Tool £0.1M Maint £1.2M IP £0.0M

VS PEERS BMW X7 £3.0M Audi Q8 £3.0M Merc GLS £5.1M

+£1.4M vs best

Insight — £4.4M — £1.4M above BMW X7. A reuse gap worth ~£1.7M to close by carrying more forward.

Regenerative Braking Software

£4.3M

ASIL C · High · reuse Medium · 28 PM · £7/veh

Brake blending control, hydraulic-electric transition, ABS/ESC coordination, one-pedal tuning, driver feel calibration.

Dev £1.3M Test £1.5M Integ £0.2M Cyber £0.0M Calib £0.1M Tool £0.1M Maint £1.0M IP £0.0M

VS PEERS BMW X7 £3.0M Audi Q8 £3.0M Merc GLS £4.9M

+£1.3M vs best

Insight — £4.3M — £1.3M above BMW X7. A reuse gap worth ~£1.6M to close by carrying more forward.

Cell Balancing Algorithms

£4.2M

ASIL C · High · reuse Medium · 28 PM · £7/veh

Active/passive balancing algorithms, balancing current control, energy routing optimisation.

Dev £1.3M Test £1.5M Integ £0.2M Cyber £0.0M Calib £0.1M Tool £0.1M Maint £1.0M IP £0.0M

VS PEERS Cayenne £4.8M

£-0.6M vs best

Insight — EV-powertrain software the German MHEV peers omit — benchmarked against the Porsche Cayenne. At £4.2M the L460 is £0.6M cheaper than the Cayenne (£4.8M), on its lower UK / Tier-1 blended rate. Heavier carry-forward would save ~£1.6M.

Camera Perception Stack

£38.9M

ASIL B · Very High · reuse Medium · 140 PM · £65/veh

Object detection/classification (DNN), lane detection, traffic sign recognition, free-space estimation, parking vision. Mono + stereo cameras.

Dev £8.0M Test £8.9M Integ £1.6M Cyber £0.6M Calib £0.5M Tool £0.9M Maint £10.2M IP £1.3M

VS PEERS BMW X7 £28.7M Audi Q8 £30.4M Merc GLS £45.3M

+£10.2M vs best*Insight — A top cost-driver: £38.9M is £10.2M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£12.2M.*

Sensor Fusion Engine

£33.7M

ASIL C · Very High · reuse Medium · 131 PM · £56/veh

Multi-modal object fusion (camera + radar + lidar if fitted), probabilistic world model, occupancy grid, track management.

Dev £7.5M Test £11.8M Integ £1.6M Cyber £0.0M Calib £0.6M Tool £0.6M Maint £9.6M IP £0.4M

VS PEERS BMW X7 £23.7M Audi Q8 £24.5M Merc GLS £39.4M

+£10.0M vs best*Insight — A top cost-driver: £33.7M is £10.0M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£12.2M.*

Path Planning Algorithms

£28.0M

ASIL C · Very High · reuse Medium · 110 PM · £47/veh

Behaviour prediction, trajectory generation, cost-function optimisation, comfort/safety trade-off. HD map integration where applicable.

Dev £6.3M Test £9.9M Integ £1.3M Cyber £0.0M Calib £0.4M Tool £0.5M Maint £8.1M IP £0.4M

VS PEERS BMW X7 £19.6M Audi Q8 £20.3M Merc GLS £32.7M

+£8.3M vs best*Insight — A top cost-driver: £28.0M is £8.3M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£10.1M.*

Highway Assist / Traffic Jam Assist

£27.4M

ASIL C · Very High · reuse Medium · 104 PM · £46/veh

Combined longitudinal + lateral control for highway driving, hands-off monitoring, lane change assist, automated speed adaptation.

Dev £5.9M Test £9.3M Integ £1.3M Cyber £0.0M Calib £0.6M Tool £0.5M Maint £7.5M IP £0.5M

VS PEERS BMW X7 £19.5M Audi Q8 £20.2M Merc GLS £32.0M

+£7.9M vs best*Insight — A top cost-driver: £27.4M is £7.9M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£9.7M.*

Control Algorithms (ACC / LKA / AEB)

£27.3M

ASIL D · High · reuse Medium · 125 PM · £45/veh

Longitudinal ACC, lateral LKA, emergency AEB, FCW, emergency steering. Control law design, tuning, gain scheduling.

Dev £5.7M Test £12.9M Integ £1.0M Cyber £0.0M Calib £0.8M Tool £0.4M Maint £6.4M IP £0.2M

VS PEERS BMW X7 £19.5M Audi Q8 £19.9M Merc GLS £31.7M

+£7.8M vs best*Insight — A top cost-driver: £27.3M is £7.8M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£9.7M.*

Driver Monitoring Software (DMS)

£14.2M

ASIL B · Very High · reuse Medium · 59 PM · £24/veh

Gaze tracking, drowsiness detection, attention estimation, hands-on-wheel detection. IR camera + CNN inference.

Dev £3.4M Test £3.4M Integ £0.5M Cyber £0.3M Calib £0.2M Tool £0.4M Maint £3.8M IP £0.7M

VS PEERS BMW X7 £10.2M Audi Q8 £10.8M Merc GLS £16.5M

+£4.0M vs best*Insight — A top cost-driver: £14.2M is £4.0M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£4.7M.*

Radar Processing Stack

£13.2M

ASIL B · Very High · reuse Medium · 65 PM · £22/veh

FMCW radar DSP, CFAR detection, Doppler processing, angle estimation, object tracking and classification.

Dev £3.7M Test £3.7M Integ £0.6M Cyber £0.0M Calib £0.3M Tool £0.4M Maint £3.5M IP £0.3M

VS PEERS BMW X7 £9.2M Audi Q8 £9.6M Merc GLS £15.4M +£4.0M vs best

Insight — A top cost-driver: £13.2M is £4.0M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£4.8M.

Automated Parking & Surround-View

£7.3M

ASIL B · High · reuse Medium · 52 PM · £12/veh

Automated parking assist (APA), remote/summon parking (RPA), 360° surround-view stitching, parking-slot detection, low-speed manoeuvre planning fusing ultrasonic + cameras.

Dev £2.4M Test £2.1M Integ £0.4M Cyber £0.0M Calib £0.2M Tool £0.2M Maint £1.9M IP £0.1M

VS PEERS BMW X7 £5.0M Audi Q8 £5.1M Merc GLS £8.6M +£2.4M vs best

Insight — £7.3M — £2.4M above BMW X7. A reuse gap worth ~£2.8M to close by carrying more forward.

Ultrasonic Processing Software

£1.7M

ASIL B · Medium · reuse Medium · 16 PM · £3/veh

Short-range object detection, parking guidance, cross-traffic alert, time-of-flight processing for 12-16 sensors.

Dev £0.6M Test £0.5M Integ £0.1M Cyber £0.0M Calib £0.0M Tool £0.1M Maint £0.4M IP £0.1M

VS PEERS BMW X7 £1.2M Audi Q8 £1.2M Merc GLS £2.0M +£0.5M vs best

Insight — £1.7M — £0.5M above BMW X7. A reuse gap worth ~£0.6M to close by carrying more forward.

IVI Operating System (Android Automotive / QNX)**£20.9M**

ASIL QM · Very High · reuse Medium · 96 PM · £35/veh

Android Automotive OS or QNX BSP integration, platform services, GPU driver optimisation, boot time optimisation, security hardening.

Dev £5.5M Test £1.8M Integ £1.2M Cyber £0.4M Calib £0.1M Tool £1.3M Maint £7.9M IP £1.8M

VS PEERS BMW X7 £15.0M Audi Q8 £16.0M Merc GLS £24.7M

+£5.9M vs best*Insight — A top cost-driver: £20.9M is £5.9M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£7.0M.***Vehicle Voice Assistant****£12.3M**

ASIL QM · Very High · reuse Medium · 39 PM · £20/veh

Wake word detection, ASR (on-device + cloud), NLU, TTS, vehicle function control, 3rd-party assistant integration (Alexa/Google).

Dev £2.2M Test £0.7M Integ £0.4M Cyber £0.2M Calib £0.1M Tool £0.5M Maint £2.8M IP £1.8M

VS PEERS BMW X7 £10.0M Audi Q8 £10.9M Merc GLS £14.2M

+£2.3M vs best*Insight — £12.3M — £2.3M above BMW X7. A reuse gap worth ~£2.7M to close by carrying more forward.***App Framework & HMI Layer****£8.5M**

ASIL QM · Very High · reuse Medium · 51 PM · £14/veh

UI framework (Qt/Flutter/React Native Automotive), instrument cluster rendering, haptics, multi-screen management, UX design system.

Dev £2.9M Test £0.8M Integ £0.4M Cyber £0.0M Calib £0.1M Tool £0.3M Maint £3.3M IP £0.7M

VS PEERS BMW X7 £5.9M Audi Q8 £6.2M Merc GLS £10.0M

+£2.6M vs best*Insight — £8.5M — £2.6M above BMW X7. A reuse gap worth ~£3.1M to close by carrying more forward.***Navigation Stack****£7.6M**

ASIL QM · High · reuse Medium · 25 PM · £13/veh

Routing engine, map rendering, offline capability, real-time traffic (HERE/TomTom), EV routing with charge planning.

Dev £1.1M Test £0.3M Integ £0.2M Cyber £0.0M Calib £0.0M Tool £0.3M Maint £1.1M IP £1.6M

VS PEERS BMW X7 £6.7M Audi Q8 £7.4M Merc GLS £9.6M

+£1.0M vs best*Insight — £7.6M — £1.0M above BMW X7. A reuse gap worth ~£1.1M to close by carrying more forward.***Telematics Control Unit (TCU) Software****£5.6M**

ASIL B · High · reuse Medium · 32 PM · £9/veh

5G/LTE modem management, emergency call (eCall), remote diagnostics, remote access, V2X readiness, OBD-II data relay.

Dev £1.5M Test £1.3M Integ £0.2M Cyber £0.1M Calib £0.1M Tool £0.2M Maint £1.4M IP £0.1M

VS PEERS BMW X7 £4.0M Audi Q8 £4.2M Merc GLS £6.5M

+£1.5M vs best*Insight — £5.6M — £1.5M above BMW X7. A reuse gap worth ~£1.8M to close by carrying more forward.***AR Head-Up Display & AR Navigation****£3.0M**

ASIL A · High · reuse Medium · 24 PM · £5/veh

Augmented-reality head-up display compositor, sensor-registered navigation/ADAS overlays, optical distortion correction, graphics rendering pipeline for the AR-HUD projector.

Dev £1.1M Test £0.5M Integ £0.2M Cyber £0.0M Calib £0.1M Tool £0.2M Maint £0.8M IP £0.1M

VS PEERS BMW X7 £2.1M Audi Q8 £2.1M Merc GLS £3.5M

+£0.9M vs best*Insight — £3.0M — £0.9M above BMW X7. A reuse gap worth ~£1.1M to close by carrying more forward.*

Bluetooth / WiFi / 5G Connectivity Stack

£2.5M

ASIL QM · High · reuse Medium · 18 PM · £4/veh

BT5.x stack (audio, phone), WiFi 6/6E AP+client, 5G SA/NSA modem driver integration, hotspot management.

Dev £0.8M Test £0.2M Integ £0.1M Cyber £0.1M Calib £0.0M Tool £0.2M Maint £0.7M IP £0.3M

VS PEERS BMW X7 £1.9M Audi Q8 £2.0M Merc GLS £2.9M

+£0.7M vs best

Insight — £2.5M — £0.7M above BMW X7. A reuse gap worth ~£0.8M to close by carrying more forward.

Premium Audio DSP & Active Noise Control

£2.5M

ASIL QM · Very High · reuse Medium · 17 PM · £4/veh

Branded 3D surround tuning (Meridian / Bowers & Wilkins / Bang & Olufsen / Burmester), active road/engine noise cancellation (ANC/RNC), active sound design, per-cabin acoustic calibration.

Dev £1.0M Test £0.2M Integ £0.1M Cyber £0.0M Calib £0.1M Tool £0.2M Maint £0.5M IP £0.3M

VS PEERS BMW X7 £1.5M Audi Q8 £1.6M Merc GLS £2.9M

+£0.9M vs best

Insight — Signature system for the L460 (Meridian 3D audio), deliberately run at Very High complexity — a brand investment, not an inefficiency. £2.5M vs £1.5M at the leanest peer (BMW X7).

Digital Key (UWB/BLE) & Secure Access

£1.9M

ASIL QM · High · reuse Medium · 13 PM · £3/veh

CCC Digital Key 3.0 phone-as-key, UWB ranging & relay-attack protection, BLE fallback, secure-element / HSM integration, key sharing & cloud provisioning backend.

Dev £0.6M Test £0.2M Integ £0.1M Cyber £0.0M Calib £0.0M Tool £0.1M Maint £0.4M IP £0.2M

VS PEERS BMW X7 £1.6M Audi Q8 £1.5M Merc GLS £2.2M

+£0.4M vs best

Insight — Competitive at £1.9M — at or near the leanest peer. Remaining reuse upside ~£0.6M.

Vehicle Motion Management (VMM)

£24.3M

ASIL C · Very High · reuse Medium · 110 PM · £41/veh

Unified torque arbitration, coordinated chassis & propulsion control, stability control, oversteer/understeer management, tyre model.

Dev £6.3M Test £8.9M Integ £1.4M Cyber £0.0M Calib £1.0M Tool £0.4M Maint £6.0M IP £0.2M

VS PEERS BMW X7 £16.7M Audi Q8 £17.1M Merc GLS £28.3M

+£7.6M vs best

Insight — A top cost-driver: £24.3M is £7.6M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£9.3M.

Chassis Control Software

£13.1M

ASIL C · High · reuse Medium · 83 PM · £22/veh

Air suspension, CDC (Continuous Damping Control), active anti-roll bars, 4WS (rear steering), hill-hold, terrain modes.

Dev £3.8M Test £4.8M Integ £0.7M Cyber £0.0M Calib £0.5M Tool £0.3M Maint £3.0M IP £0.1M

VS PEERS BMW X7 £9.1M Audi Q8 £9.3M Merc GLS £15.2M

+£4.0M vs best

Insight — A top cost-driver: £13.1M is £4.0M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£4.9M.

Zonal Architecture Software

£13.1M

ASIL B · Very High · reuse Medium · 70 PM · £22/veh

Zone controller software, power distribution management, ECU consolidation logic, 100BASE-T1 Ethernet backbone management.

Dev £4.0M Test £3.4M Integ £0.8M Cyber £0.3M Calib £0.2M Tool £0.4M Maint £3.8M IP £0.2M

VS PEERS BMW X7 £8.8M Audi Q8 £9.1M Merc GLS £15.3M

+£4.3M vs best

Insight — A top cost-driver: £13.1M is £4.3M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£5.1M.

Predictive Active Suspension & Body Control

£12.4M

ASIL C · Very High · reuse Medium · 58 PM · £21/veh

Camera-fed road-surface preview, 48V active anti-roll / active body control (e.g. Mercedes E-Active Body Control, BMW Executive Drive Pro, JLR Dynamic Response Pro), coordinated air-spring + adaptive-damper + rear-axle-steer control on top of baseline chassis control.

Dev £3.3M Test £4.6M Integ £0.7M Cyber £0.0M Calib £0.5M Tool £0.3M Maint £2.9M IP £0.1M

VS PEERS BMW X7 £8.5M Audi Q8 £8.7M Merc GLS £14.4M

+£3.9M vs best

Insight — A top cost-driver: £12.4M is £3.9M above the leanest peer (BMW X7). The gap is software reuse — carrying this module forward (Heavy) recovers ~£4.7M.

Body Control Software

£4.9M

ASIL B · Medium · reuse Medium · 49 PM · £8/veh

Lighting control, power windows/mirrors/sunroof, door locks, wiper logic, ambient lighting, comfort features state machines.

Dev £1.9M Test £1.3M Integ £0.3M Cyber £0.0M Calib £0.1M Tool £0.1M Maint £1.2M IP £0.1M

VS PEERS BMW X7 £3.4M Audi Q8 £3.4M Merc GLS £5.7M

+£1.6M vs best

Insight — £4.9M — £1.6M above BMW X7. A reuse gap worth ~£1.9M to close by carrying more forward.

Gateway ECU Software

£4.2M

ASIL B · High · reuse Medium · 30 PM · £7/veh

CAN/LIN/FlexRay/Ethernet routing, signal translation, diagnostic gateway, firewall, network management master.

Dev £1.4M Test £1.1M Integ £0.2M Cyber £0.1M Calib £0.1M Tool £0.2M Maint £1.1M IP £0.1M

VS PEERS BMW X7 £2.9M Audi Q8 £3.0M Merc GLS £4.9M

+£1.3M vs best

Insight — £4.2M — £1.3M above BMW X7. A reuse gap worth ~£1.6M to close by carrying more forward.

Climate & Thermal Comfort Control

£2.0M

ASIL A · Medium · reuse Medium · 21 PM · £3/veh

Multi-zone HVAC control, heat-pump coordination, cabin air purification/ionisation, pre-conditioning, humidity & fogging management, seat/steering-wheel comfort thermal loops.

Dev £0.8M Test £0.3M Integ £0.1M Cyber £0.0M Calib £0.1M Tool £0.1M Maint £0.5M IP £0.0M

VS PEERS BMW X7 £1.4M Audi Q8 £1.4M Merc GLS £2.3M +£0.6M vs best

Insight — £2.0M — £0.6M above BMW X7. A reuse gap worth ~£0.8M to close by carrying more forward.

AUTOSAR Adaptive Platform**£11.3M**

ASIL B · High · reuse Medium · 65 PM · £19/veh

ara::com service-oriented communication, execution management, update management (UCM), PHM, crypto API, DDS integration.

Dev £2.9M Test £2.5M Integ £0.6M Cyber £0.2M Calib £0.1M Tool £0.9M Maint £2.8M IP £0.9M

VS PEERS BMW X7 £8.1M Audi Q8 £5.3M Merc GLS £13.1M

+£6.0M vs best*Insight — The L460 develops this in-house while Audi runs it as shared VW-Group middleware (Platform reuse). That is why the L460 sits £6.0M above Audi Q8. Adopting a shared-platform stack is the targeted fix here (~£3.8M).***AUTOSAR Classic BSW & Integration****£10.5M**

ASIL C · High · reuse Medium · 62 PM · £18/veh

AUTOSAR Classic BSW configuration (Vector DaVinci, ETAS ISOLAR), OS (OSEK/VDX), COM stack, run-time environment (RTE) integration.

Dev £2.8M Test £3.1M Integ £0.6M Cyber £0.0M Calib £0.1M Tool £1.1M Maint £1.8M IP £1.0M

VS PEERS BMW X7 £7.9M Audi Q8 £5.3M Merc GLS £12.1M

+£5.2M vs best*Insight — The L460 develops this in-house while Audi runs it as shared VW-Group middleware (Platform reuse). That is why the L460 sits £5.2M above Audi Q8. Adopting a shared-platform stack is the targeted fix here (~£3.2M).***CAN / LIN / FlexRay / Ethernet Communication Stacks****£2.5M**

ASIL B · Medium · reuse Medium · 22 PM · £4/veh

AUTOSAR ComStack configuration, CAN-XL support, Automotive Ethernet (10/100BASE-T1), signal routing, PDU mapping, SOME/IP.

Dev £0.8M Test £0.6M Integ £0.1M Cyber £0.0M Calib £0.0M Tool £0.2M Maint £0.5M IP £0.2M

VS PEERS BMW X7 £1.8M Audi Q8 £1.2M Merc GLS £2.8M

+£1.3M vs best*Insight — The L460 develops this in-house while Audi runs it as shared VW-Group middleware (Platform reuse). That is why the L460 sits £1.3M above Audi Q8. Adopting a shared-platform stack is the targeted fix here (~£0.8M).***RTOS & OS Porting****£2.2M**

ASIL C · Medium · reuse Medium · 17 PM · £4/veh

OSEK/VDX, FreeRTOS, or commercial RTOS BSP integration, task scheduling optimisation, interrupt management, memory protection.

Dev £0.6M Test £0.7M Integ £0.1M Cyber £0.0M Calib £0.0M Tool £0.2M Maint £0.3M IP £0.3M

VS PEERS BMW X7 £1.7M Audi Q8 £1.2M Merc GLS £2.6M

+£1.0M vs best*Insight — The L460 develops this in-house while Audi runs it as shared VW-Group middleware (Platform reuse). That is why the L460 sits £1.0M above Audi Q8. Adopting a shared-platform stack is the targeted fix here (~£0.7M).***Diagnostics (UDS / OBD-II)****£2.2M**

ASIL A · Medium · reuse Medium · 20 PM · £4/veh

UDS (ISO 14229), OBD-II (SAE J1979), DTC definition, diagnostic service implementation, DEXT generation, coding routines.

Dev £0.8M Test £0.3M Integ £0.1M Cyber £0.0M Calib £0.0M Tool £0.1M Maint £0.5M IP £0.1M

VS PEERS BMW X7 £1.6M Audi Q8 £1.7M Merc GLS £2.6M

+£0.6M vs best*Insight — £2.2M — £0.6M above BMW X7. A reuse gap worth ~£0.7M to close by carrying more forward.***Time Synchronisation & Network Management****£0.9M**

ASIL A · Low · reuse Medium · 11 PM · £2/veh

gPTP (IEEE 802.1AS) for Ethernet time sync, CAN NM master, sleep/wake management, partial networking.

Dev £0.4M Test £0.2M Integ £0.0M Cyber £0.0M Calib £0.0M Tool £0.1M Maint £0.2M IP £0.0M

VS PEERS BMW X7 £0.6M Audi Q8 £0.7M Merc GLS £1.1M

+£0.3M vs best*Insight — Competitive at £0.9M — at or near the leanest peer. Remaining reuse upside ~£0.3M.*

Secure OTA Software Update Framework

£4.7M

ASIL B · High · reuse Medium · 22 PM · £8/veh

Delta update generation, signature verification, rollback protection, update orchestration, bandwidth management.

Dev £1.0M Test £0.9M Integ £0.2M Cyber £0.1M Calib £0.0M Tool £0.2M Maint £1.1M IP £0.3M

VS PEERS BMW X7 £3.6M Audi Q8 £3.9M Merc GLS £5.5M

+£1.1M vs best*Insight — £4.7M — £1.1M above BMW X7. A reuse gap worth ~£1.3M to close by carrying more forward.*

Secure Boot & Root of Trust

£4.0M

ASIL B · High · reuse Medium · 22 PM · £7/veh

Hardware Security Module (HSM) integration, key provisioning, boot chain verification, anti-rollback, attestation.

Dev £1.0M Test £1.0M Integ £0.2M Cyber £0.1M Calib £0.0M Tool £0.2M Maint £0.9M IP £0.2M

VS PEERS BMW X7 £2.9M Audi Q8 £3.0M Merc GLS £4.6M

+£1.1M vs best*Insight — £4.0M — £1.1M above BMW X7. A reuse gap worth ~£1.3M to close by carrying more forward.*

Intrusion Detection System (IDS)

£3.6M

ASIL QM · High · reuse Medium · 17 PM · £6/veh

In-vehicle network anomaly detection, CAN message monitoring, rate-limiting, VSOC integration, event reporting to cloud.

Dev £0.8M Test £0.3M Integ £0.1M Cyber £0.1M Calib £0.0M Tool £0.3M Maint £1.0M IP £0.4M

VS PEERS BMW X7 £2.8M Audi Q8 £3.0M Merc GLS £4.2M

+£0.8M vs best*Insight — £3.6M — £0.8M above BMW X7. A reuse gap worth ~£0.9M to close by carrying more forward.*

Key Management System

£3.3M

ASIL B · High · reuse Medium · 16 PM · £6/veh

Certificate lifecycle management, PKI integration, key derivation, secure key storage, provisioning infrastructure.

Dev £0.7M Test £0.7M Integ £0.1M Cyber £0.1M Calib £0.0M Tool £0.2M Maint £0.7M IP £0.2M

VS PEERS BMW X7 £2.5M Audi Q8 £2.7M Merc GLS £3.8M

+£0.8M vs best*Insight — £3.3M — £0.8M above BMW X7. A reuse gap worth ~£0.9M to close by carrying more forward.*

Encryption Modules & Crypto Stack

£2.1M

ASIL B · Medium · reuse Medium · 16 PM · £3/veh

AES-256, RSA-2048, ECC, TLS 1.3 for V2X/cloud, AUTOSAR Crypto Stack, hardware crypto acceleration.

Dev £0.6M Test £0.6M Integ £0.1M Cyber £0.0M Calib £0.0M Tool £0.1M Maint £0.5M IP £0.1M

VS PEERS BMW X7 £1.5M Audi Q8 £1.5M Merc GLS £2.4M

+£0.6M vs best*Insight — £2.1M — £0.6M above BMW X7. A reuse gap worth ~£0.7M to close by carrying more forward.*

Cloud Backend Services**£18.1M**

ASIL QM · Very High · reuse Medium · 39 PM · £30/veh

Vehicle connectivity backend, API gateway, device shadow, remote command, data lake, microservices architecture (AWS/Azure/GCP).

Dev £2.2M Test £0.7M Integ £0.4M Cyber £0.2M Calib £0.1M Tool £0.4M Maint £3.6M IP £1.0M

VS PEERS BMW X7 £15.6M Audi Q8 £17.2M Merc GLS £20.8M

+£2.5M vs best*Insight — £18.1M — £2.5M above BMW X7. A reuse gap worth ~£2.9M to close by carrying more forward.***OTA Update Manager (Vehicle-side)****£6.7M**

ASIL B · High · reuse Medium · 32 PM · £11/veh

Vehicle-side update campaign execution, ECU coordination, rollback, consent management, network condition handling.

Dev £1.5M Test £1.3M Integ £0.3M Cyber £0.1M Calib £0.1M Tool £0.2M Maint £1.4M IP £0.4M

VS PEERS BMW X7 £5.1M Audi Q8 £5.4M Merc GLS £7.7M

+£1.5M vs best*Insight — £6.7M — £1.5M above BMW X7. A reuse gap worth ~£1.9M to close by carrying more forward.***Data Pipeline & Telemetry****£5.5M**

ASIL QM · High · reuse Medium · 18 PM · £9/veh

In-vehicle data collection agent, edge pre-processing, telemetry streaming, data lake ingestion, GDPR/data governance.

Dev £0.8M Test £0.2M Integ £0.1M Cyber £0.1M Calib £0.0M Tool £0.2M Maint £0.9M IP £0.5M

VS PEERS BMW X7 £4.7M Audi Q8 £5.2M Merc GLS £6.3M

+£0.8M vs best*Insight — £5.5M — £0.8M above BMW X7. A reuse gap worth ~£0.9M to close by carrying more forward.***Fleet Management & Analytics Software****£3.3M**

ASIL QM · Medium · reuse Medium · 12 PM · £6/veh

Fleet health dashboard, predictive maintenance, usage analytics, dealer portal, subscription management, over-the-air campaign tools.

Dev £0.5M Test £0.1M Integ £0.1M Cyber £0.0M Calib £0.0M Tool £0.1M Maint £0.6M IP £0.4M

VS PEERS BMW X7 £2.9M Audi Q8 £3.2M Merc GLS £3.8M

+£0.4M vs best*Insight — £3.3M — £0.4M above BMW X7. A reuse gap worth ~£0.5M to close by carrying more forward.*

C Where the £145M sits, by module

The MHEV optimisation (reuse "Medium to Heavy" + shared middleware) decomposed to the modules that contribute most. Ranked by module-level saving.

MODULE	DOMAIN	L460 COST	SAVING (HEAVY)
Camera Perception Stack	B	£38.9M	12.2M
Sensor Fusion Engine	B	£33.7M	12.2M
Path Planning Algorithms	B	£28.0M	£10.1M
Control Algorithms (ACC / LKA / AEB)	B	£27.3M	£9.7M
Highway Assist / Traffic Jam Assist	B	£27.4M	£9.7M
Vehicle Motion Management (VMM)	D	£24.3M	£9.3M
IVI Operating System (Android Automotive / QNX)	C	£20.9M	£7.0M
Zonal Architecture Software	D	£13.1M	£5.1M
Chassis Control Software	D	£13.1M	£4.9M
Radar Processing Stack	B	£13.2M	£4.8M
Driver Monitoring Software (DMS)	B	£14.2M	£4.7M
Predictive Active Suspension & Body Control	D	£12.4M	£4.7M
AUTOSAR Adaptive Platform	E	£11.3M	£3.8M
AUTOSAR Classic BSW & Integration	E	£10.5M	£3.2M
App Framework & HMI Layer	C	£8.5M	£3.1M
Cloud Backend Services	G	£18.1M	£2.9M

Reuse alone recovers ~£138.6M on the MHEV programme; shared-platform middleware adds ~£6.5M. ADAS (Category B) carries roughly half the opportunity — it is the largest domain, so the reuse rate compounds there most.

D Assumptions & methodology

Every figure is live output from the CostVision should-cost engine (49-module catalogue, ISO 26262 / 21434). Nothing is hand-entered.

Model class & accuracy

Bottom-up parametric should-cost; validated envelope $\pm 25\text{--}35\%$ (back-test MAPE 24.9%). Absolute numbers are estimates; comparisons are consistent (same engine + rate library).

Comparison basis

MHEV modules priced for L460 (UK/Tier-1/75k/Medium) vs BMW X7, Audi Q8, Mercedes GLS. The 7 EV-powertrain modules — which the German MHEVs omit — priced in the BEV configuration vs the Porsche Cayenne.

Per-module cost

Total programme should-cost = development + testing + integration + cybersecurity + calibration + toolchain (NRE) plus maintenance + cloud + IP licence (lifecycle), over programme life.

Reuse factors (× dev effort)

Fresh 1.00 · Light 0.82 · Medium 0.60 · Heavy 0.35 · Platform 0.14. Module-level "saving" is the effect of moving that module from the L460 baseline reuse to "Heavy".

Signature software

Brand-DNA overrides held constant (Meridian on L460; MBUX/voice on GLS; Porsche 4D chassis on Cayenne; UWB key on X7).

Propulsion availability

Verified July 2026: X7/Q8/GLS are MHEV-only in production; Cayenne offers ICE/PHEV/BEV; Range Rover Electric is launching. No fabricated variants are benchmarked.

Currency & scope

All figures GBP. Embedded & connected vehicle software only; excludes manufacturing, hardware and non-software engineering.

This report supports target-setting and supplier negotiation; it is not an audited actual. Engine: CostVision sw-should-cost.ts, rate library v1.0, July 2026.